Danmarks Tekniske Universitet

Take-home exam: December 19, 2012.

Course name: Network Optimization **Course no:** 42115

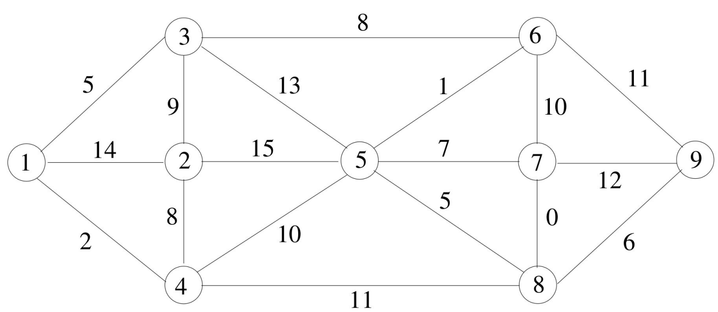
Rules: You are allowed to use your books, notes, slides, calculator, LP-solver etc. You are *not* allowed to get help from other people. You *must* fill in and sign the front page (last page in this document).

Assignment: The assignment consists of 15 problems. In some of the problems it is written what an answer is expected to contain. Notice that all figures can be found at the last pages of this assignment. You can print the figures, and use them in your answer by drawing on the figure.

Points: Small problems give 4 points, medium problems give 6 points, large problems 10 points, and xlarge problems give 14 points. A total of 100 points can be obtained.

Minimum spanning tree

Given the following graph $G = (V,E)$

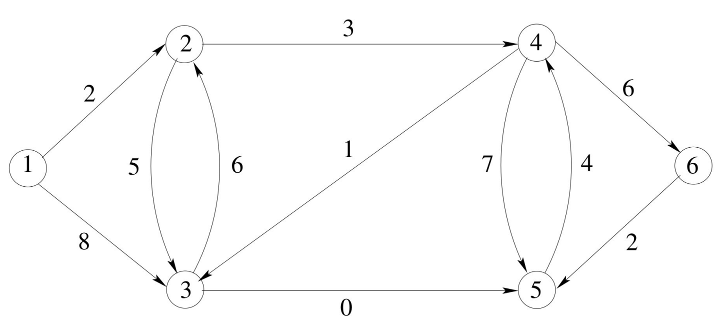


Problem 1: (small) Find the minimum spanning tree of G using Kruskal's algorithm ■

Problem 2: (small) Find the minimum spanning tree of G using Prim's algorithm ■

It is sufficient to indicate the chosen edges on the graph and to give each edge a sequence number (in which order they have been considered).

Shortest path



Problem 3: (small) Use Dijkstra's algorithm to solve the above shortest path problem starting from node 1. ■

It is sufficient to indicate the chosen edges in a shortest path tree on the graph and to give each edge a sequence number (in which order they have been considered). Moreover, the distance to each node should be indicated on the node.

Shortest path tree

Problem 4: (large) Are the following statements correct? (Prove or disprove each statement. To disprove a statement it is sufficient to show an example.)

a) The shortest path tree is unique when all edges have distinct weights.

b) In a directed graph with positive edge weights the shortest paths do not change when directed edges are replaced with undirected edges.

c) In a shortest path problem, if all edge weights are increased by adding a constant $k > 0$ then the cost of a shortest path is also increased by a multiple of k.

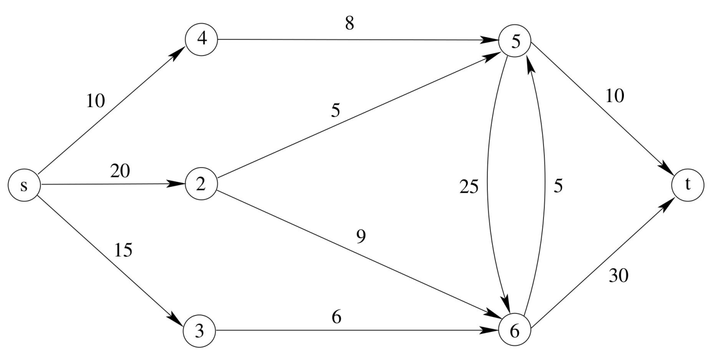
d) Among all shortest paths in $G = (V, E)$ the Dijkstra algorithm finds the shortest path with fewest edges.

e) Among all shortest paths in $G = (V, E)$ the Bellman-Ford algorithm finds the shortest path with fewest edges.

■

Maximum flow problem

Consider the maximum flow problem with source s and sink t

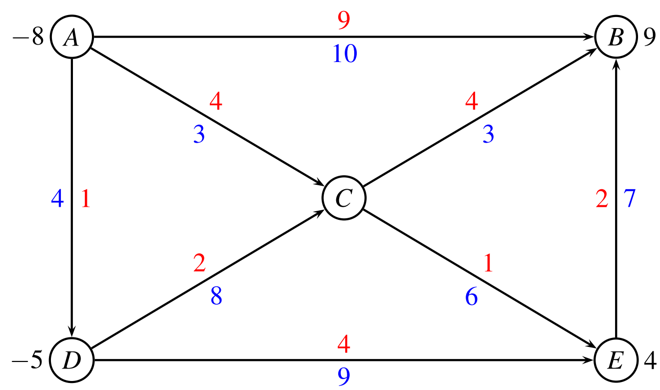


Problem 5: (medium) Solve the max flow problem using an augmenting path algorithm. In each iteration let the augmenting path be the shortest (measured in number of edges) path from s to t in the residual graph. In each iteration write down which augmenting path was used. ■

Problem 6: (small) Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i in every node is positive for a sink, and negative for a source.



Problem 7: (small) Find first a legal start solution by solving an appropriate max-flow problem. ■

It is sufficient to draw the max-flow problem and to show the solution. No details about how the max-flow problem is solved are needed.

Problem 8: (large) Use the augmenting cycle algorithm on the above network. In each iteration, write down which augmenting cycle was used. ■

No details about how you found an augmenting cycle are needed.

Text formatting

The well-known document processing program LATEX uses an optimization procedure to decompose a paragraph into several lines so that when lines are left- and right-adjusted, the appearance of the paragraph will be the most attractive. Suppose that a paragraph consists of n words and that each word is assigned a sequence number. Let $c_{ij} > 0$ denote the attractiveness of a line if it begins with the word i and ends with word $j-1$. Bigger values of c_{ij} are better. The program LATEX uses formulas to calculate the value of each c_{ij}, and we can assume they are given.

Problem 9: (large) Given the c_{ij} values, show how to formulate the problem of decomposing the paragraph into several lines of text in order to maximize the total attractiveness (of all lines) as a shortest path problem. ■

Problem 10: (small) Does the graph have any negative cycles? ■

Problem 11: (medium) Propose an algorithm for solving the above problem. Try to find an algorithm with the best possible running time (in big-Oh notation). ■

Hotel planning

A hotel has received a number of requests for the following week as shown in the following table

start\end	Wed	Thu	Fri
Mon	2	3	-
Tue	4	3	5
Wed	-	4	3

Each entry in the table gives the number of guests requesting a room from the "start" day to the "end" day. A dash indicates that no requests have been given. A stay may not be split.

The hotel gets 2$ for a 1-day stay, 5$ for a 2-day stay, and 8$ for a 3-day stay. The hotel wants to maximize the revenue. However, only the following number of rooms is available each of the nights:

Mon-Tue	Tue-Wed	Wed-Thu	Thu-Fri
5	6	8	7

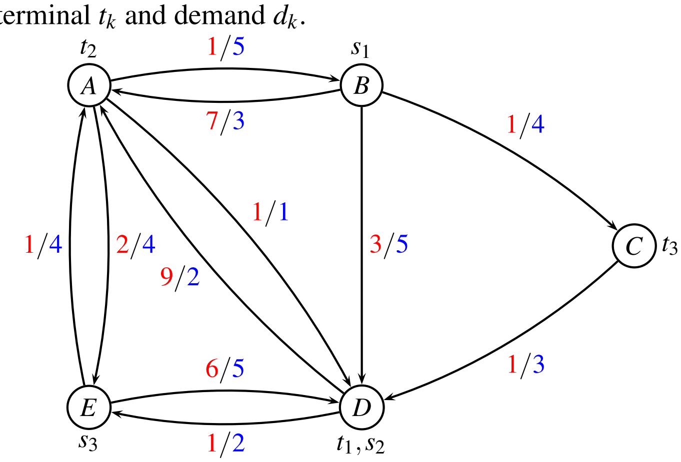
Problem 12: (xlarge) Formulate the problem as a minimum cost flow problem by drawing a graph $G = (V, E)$ where each edge (i, j) has a cost c_{ij}, and an upper bound on the flow u_{ij}. Moreover, every node i has a demand b_i which is negative for a source, and positive for a sink. ■

Hint: See application 9.4 in Ahuja, Magnanti, Orlin "Network Flows". (*Available on campusnet*).

Problem 13: (medium) Solve the problem with CPLEX or any other solver. Enclose a copy of the MIP model, and report the optimal solution. ■

Multicommodity flow

Consider the following unsplittable multicommodity flow problem where each edge has a cost and a capacity c_{ij}/u_{ij}. There are three commodities as indicated in the table, each commodity k having source s_k, terminal t_k and demand d_k.



k	s_k	t_k	d_k
1	B	D	1
2	D	A	2
3	E	C	2

A path formulation of the problem is given in the last page of the appendix.

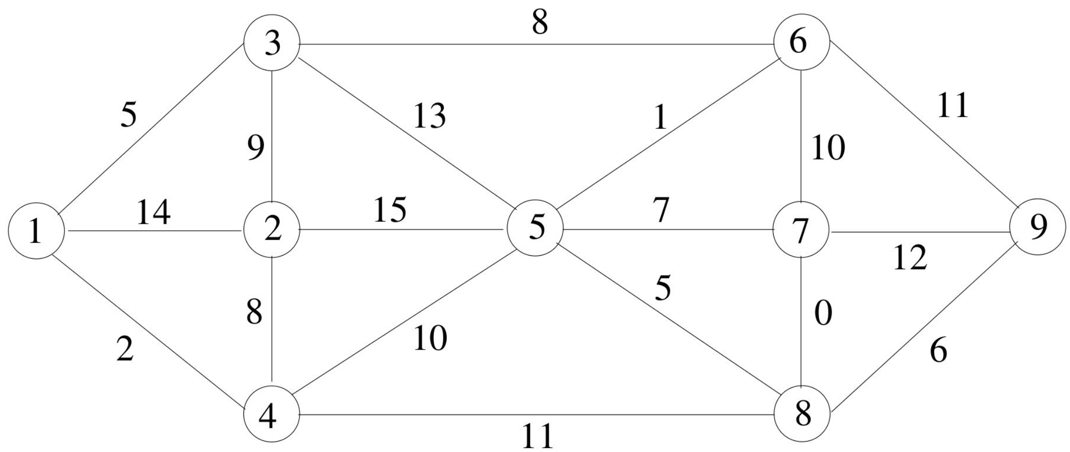
Problem 14: (small) Fill in the objective value in the path formulation, and solve the problem to LP-optimality and IP-optimality. Notice that the number of commodities flown is handled in the last constraints, as opposed to the project assignment. ■

Problem 15: (large) Run column generation starting from columns corresponding to the variables x_1, x_5, x_7. ■

Hint: In the last question you can add a constraint $x_i \leq 0$ to all columns not being used. It is sufficient to show the input (written as an LP-problem) and the primal/dual solution in each iteration. You may use CPLEX or any other LP-solve. When reporting the input it should be written as a normal LP-problem.

Appendix

The figures on the following pages can be used when solving the problem. It is allowed to write by hand on the figures in your answer.



8
3
6
5
13
1
11
9
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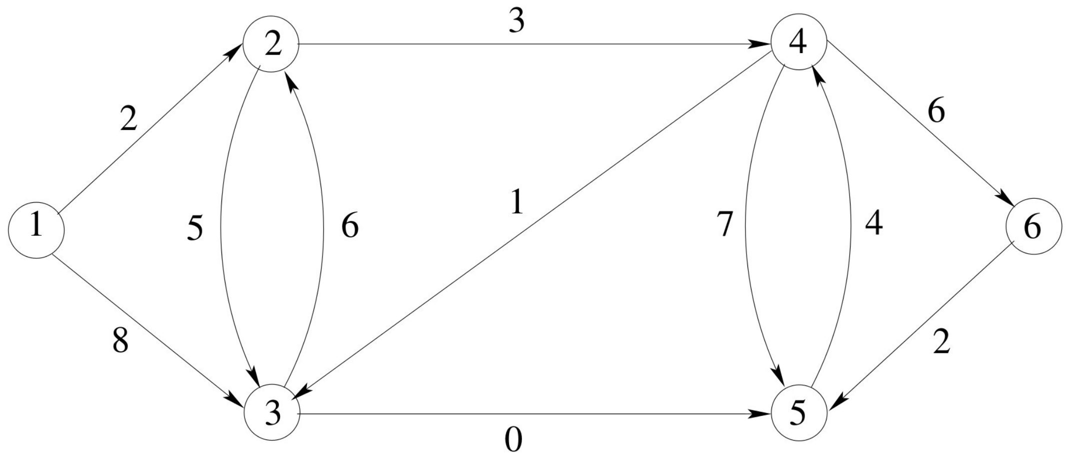

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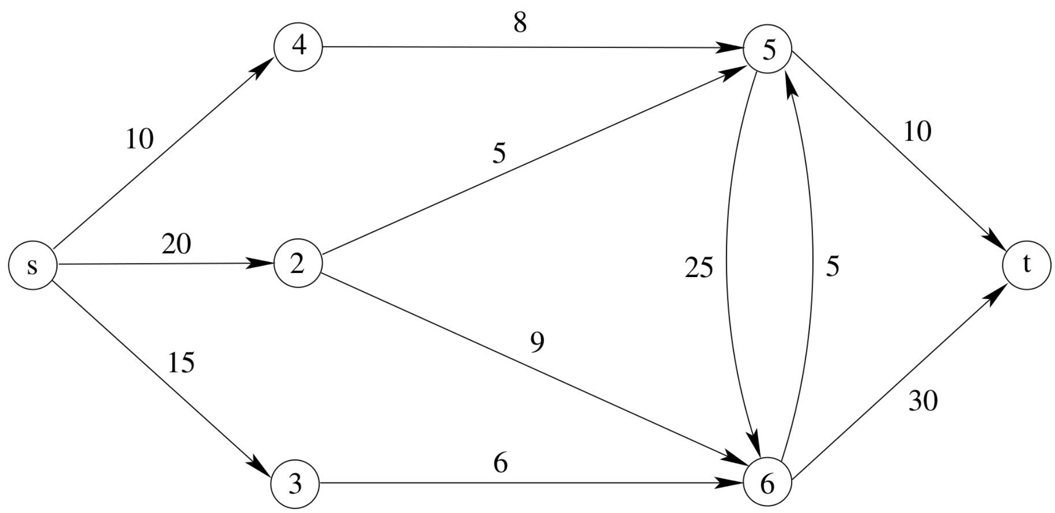
s
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3
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t
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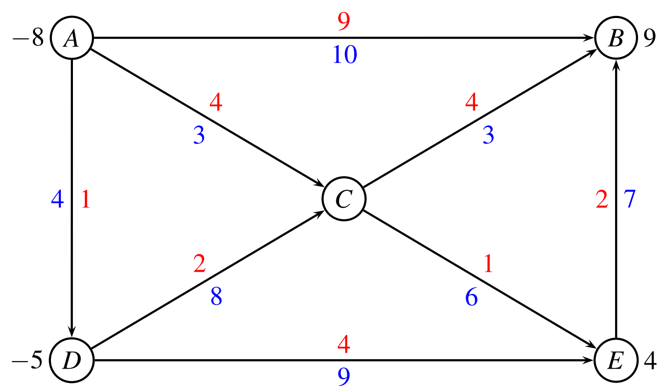
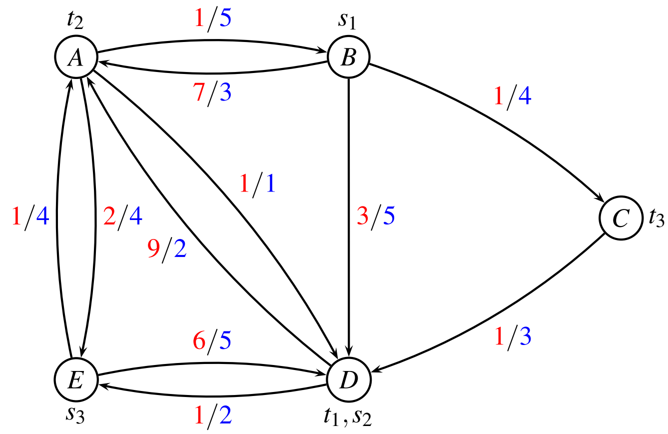
20
15
8
5
9
6
25
5
10
30



A
B
C
D
E
−8
9
−5
4
9
10
4
3
4
3
4
1
2
7
2
8
1
6
4
9



t_2
A
$1/5$
s_1
B
$7/3$
$1/4$
$1/1$
$1/4$
$2/4$
$9/2$
$3/5$
C
t_3
$6/5$
$1/3$
E
D
s_3
$1/2$
t_1, s_2

```
minimize
 + x1+ x2+ x3+ x4+ x5+ x6+ x7+ x8
subject to
AB:                   +x7+x8 <= 5
AD: +x1                      <= 1
AE:    +x2                   <= 4
BA: +x1+x2                   <= 3
BC:       +x3         +x7+x8 <= 4
BD:          +x4             <= 5
CD:       +x3                <= 3
DA:             +x5      +x8 <= 2
DE:                +x6       <= 2
EA:                +x6+x7    <= 4
ED:    +x2               +x8 <= 5

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 2
K3: +x7+x8 >= 2

end
```

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Name	study number

I hereby declare that I have worked alone on the project assignment and not discussed it with anybody else.

Date Signature